

Unit

1

Lesson

2



Spark The Challenge!

The Magic of Vibrations.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify how sounds are made and how they move from place to place.
- Describe how vibrations cause sound and the shape of sound waves.
- Compare how sound travels through solids, liquids, and gases.
- Explain how understanding sound can help people in real life.
- Understand basic cause and effect.

Challenge questions of the lesson

1. What would happen if someone rang a bell in space where there is no air?
2. How do different vibrations create different sounds, and what do their sound waves look like?
3. Who would hear a sound faster—a dolphin underwater or a person in the air—and why?
4. How can understanding how sound works help people like doctors or engineers in their jobs?

SEL Relation

- Responsible Decision-Making: Thinking critically and applying scientific reasoning to understand how sound travels.
- Self-Awareness: Exploring how sounds relate to emotions and creativity (e.g., music).
- Self-Management: Engaging in hands-on learning with focus and curiosity.
- Social Awareness: Understanding and appreciating how animals and humans experience the world differently.
- Responsible Decision-Making: Recognizing how knowledge can be used to solve real-life problems.
- Social Awareness: Appreciating how science supports others' well-being.
- Self-Awareness: Noticing how actions change outcomes and affect experiences (e.g., sound in music).
- Self-Management: Practicing patience and control while experimenting and learning.

Engage



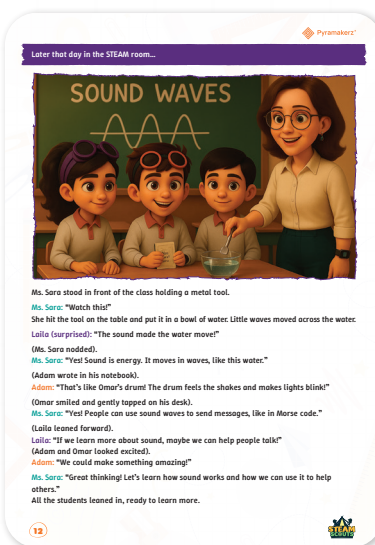
Teacher's Role:

- Introduce the story by reminding students of Omar and the idea of building something to help him communicate.
- Read the story aloud with expression, especially emphasizing curiosity and discovery.
- Highlight key phrases ("Sound and Vibration: More Than Just Noise") to spark wonder.
- Ask questions during and after reading to connect emotions and concepts: "Why did the lights blink when Omar hit the drum?"

Expected From Students:

- Listen carefully and identify how sound and movement are connected.
- Ask questions and make observations about what sound can do.
- Predict how sound might be useful for communication.
- Share personal ideas and experiences in classroom discussion.

Parts of the books included.



Story



- **Begin by asking:**
 - “Have you ever been to a science fair? What kinds of projects do you think people might bring?”
 - Encourage a few students to share ideas (lights, robots, sounds, posters).
- **Then say:**
 - “Today, Adam and Lila are visiting a science fair too—and guess what? Omar has something surprising at his table. Let’s read and see what it is!”
- **Preview the story’s key question:**
 - “How can sound help people communicate?”

Possible Strategies for Implementation:

1. Sound Warm-Up Activity

Before reading, have students close their eyes. Clap once. Ask:
“What did you feel? What did you hear? Can sound move?”

2. Guided Story Reading with Expression

- Read with changes in tone for each character (e.g., curious, excited, surprised).
- Emphasize vocabulary like “shakes,” “vibration,” “waves,” and “energy.”

3. Prediction Prompts

- Pause after:
 - “The lights blinked with every beat.”
 - Ask: “Why do you think that happened?”
 - Let students guess before reading Omar’s explanation.

4. Visual and Tactile Connections

- Bring in a real drum or use a plastic container. Tap it and ask:
 - “What do you see? Can you feel the sound moving?”
 - Help them connect this to the water demonstration later.

5. Post-Story Reflection Prompts

- Ask:
 - “Why do you think Omar smiled and gave a thumbs-up?”
 - “What did Adam and Lila want to build?”
 - “What problem are they trying to solve?”

Let's Think!



Objective:

Students will explore their sensory observations of sound, reflect on sound as energy, and begin to understand that sound travels through movement—even if it can't be seen.

Answer:

Clap your hands once

What did you feel in your hands? What did you hear?

- A tiny shake or tingle.
- A soft vibration or pulse.

Can you see the sound?

- No, but I can feel or hear it.
- It moves, but I can't see it.

Where do you think the BAM sound comes from?

- From my hands shaking.
- From the air around us.
- It moves from my hands to someone's ears.

Possible Strategies for implementation:

1. Clap + Close Eyes.

- Ask students to clap, then close their eyes and describe the sensation.
- Ask: "Did you feel it just in your hands or somewhere else?"

2. Partner Discussion.

- Pair students and let them share what they felt and heard.
- Encourage each pair to guess where the sound goes.

3. Visual Connection.

- Show an image of a sound wave and ask:
 - "If we could see sound, what might it look like?"

4. Use Hand Motions.

- Have students pretend to "push" air with their hands after clapping.
- Reinforce that sound is caused by movement and travels through the air.

Explorer's Toolkit



Teacher's Role:

- Guide students through each experiment in the DIY Lab.
- Explain that they are testing how sound moves through solids, liquids, and air.
- Ensure students follow safety steps (no heads in water).
- Pause between each part to let students record their observations.

Expected From Students:

- Test how sound travels through different materials (spoons, water, air).
- Record what they hear or feel.
- Reflect on how sound travels in different states of matter.

Parts of the books included.



DIY Lab

Objective:

Students will investigate how sound travels through different materials—solid, liquid, and gas—by observing vibrations and listening carefully in hands-on experiments.

Materials Needed:

- 2 or 3 metal spoons.
- String.
- Bowl of water.
- Pencil or stick.
- Optional: soft cloth for quiet space.
- Teacher resource video: Watch: How Sound Travels.

Activity Instructions

1. Spoons in Solid Contact (Air + Metal)

- Tie spoons together and let them hang from strings wrapped around students' fingers.
- Students place their fingers gently in their ears and a partner taps the spoons.
- Ask: "What do you hear?" "Where do you feel it?"

2. Spoons in Liquid (Water)

- Place one spoon in a bowl of water. Tap it gently with a pencil.
- Students bring their ears close to the water (without touching it).
- Ask: "Does the sound change in water?"

3. Spoon in Air (Gas)

- Hold the spoon in the air and tap gently. Ask students to listen.
- Discuss how sound is different in air compared to solids or liquids.

Possible Strategies for Implementation:

- **Predict Before Testing:** Ask students: "Where do you think sound will travel best—solid, water, or air?" Write predictions on the board.
- **Rotate in Small Groups:** Have students move through each test in small, teacher-supervised stations.
- **Prompt Reflection:** After each step, ask: "Was that sound louder or softer? Did you feel it or just hear it?"
- **Use Anchor Charts:** Make a chart labeled Solid–Liquid–Gas. Record group findings visually.

Expected Answers

1. Spoons in the air (Solid through string & air):

What do you hear?

- A soft or dull sound
- Light tapping or clinking
- Not very loud

2. Spoons in water (Liquid):

What do you hear?

- A muffled or quieter sound
- Some echo or bubbling feeling
- Sound feels like it moves slower

3. Spoon tapped in open air (Gas):

What do you hear?

- A clearer or louder tapping sound
- Sound travels fast and is easy to hear
- Possibly sharper than water sound

How Does Sound Move?

Solid	Liquid	Gas
		
The sound is loud and clear.	The sound is soft.	The sound is quiet.

Explain



Teacher Role:

- Introduce the video by saying: “Let’s see how our voice makes sound and how sound travels to our ears.”
- Pause the video at key moments to clarify: vibrations, vocal cords, air.
- Ask comprehension questions during and after viewing.

Expected From Students:

- Watch attentively.
- Note down how sound is made, and why it needs air.
- Share surprising facts or ask new questions.

Parts of the books included.



Screen Lessons

Let’s watch a video to see:

1. How your voice makes sound?
2. Why sound needs air to travel?
3. How ears catch sound waves?

<https://www.youtube.com/watch?v=gdGyvGPZ1G0>

Figure 10: A girl hitting three spoons together to hear sound traveling through the air.



Screen Lessons

Objective:

Students will understand how sound is made by vibrations, how it travels through air, and how ears receive and interpret sound waves.

Video Link: <https://www.youtube.com/watch?v=gdGyvGPZ1G0>



Possible Strategies for implementation:

Before Watching: Activate Curiosity

- Ask students:
 - “Where do you think your voice comes from?”
 - “Can sound travel in space?”
 - “Why do astronauts wear special radios?”

1. Guided Viewing

- Pause the video at each key question (how voice works, how sound travels, how ears work).
- Use gestures to reinforce concepts (e.g., wiggle fingers to show sound waves).

2. Think-Pair-Share

- After the video, let students share what surprised them or what was hard to understand.
- Encourage them to use words like vibration, air, and wave.

3. Quick Sketch Challenge

- Ask students to draw what they think sound waves look like or how the ear catches sound.
- Let them label their sketches using words from the video.

Pre-Watching: Set the Purpose

Goals: Spark curiosity and activate prior knowledge.

- Ask students:
 - “What part of your body helps you make sound?”
 - “Can we talk in space? Why or why not?”
 - “What would happen if there were no air around us?”
- Invite predictions:
 - “What do you think sound needs to travel?”
 - “Where do you think sound goes after you make it?”

During Watching: Active Engagement

Goals: Focus students on key concepts.

- Pause after major concepts to ask quick questions like:
 - “What did you notice when the vocal cords vibrated?”
 - “What helped the sound get from the mouth to the ear?”
- Use hand motions or classroom models to demonstrate:
 - Vibrations (shake fingers),
 - Sound waves (wavy arm movement),
 - Air as a carrier (fan motion).

After Watching: Discussion & Reflection

Goals: Reinforce learning and check for understanding.

- Review the video’s main points using the questions provided in the activity:

Answer Key to Video Questions

1. How your voice makes sound?

Your vocal cords vibrate when air passes through them. These vibrations make sound.

2. Why sound needs air to travel?

Sound is a vibration that needs something to move through. It uses air (or other matter) to travel. No air = no sound!

How ears catch sound waves?

Your ears collect the sound waves, which go inside and shake small parts like the eardrum.

Your brain turns these into sounds you understand.

Elaborate and Evaluate



Teacher's Role:

- Guide students through Apprentice and Builder activities.
- Help students connect experiments to real-life sound movement.
- Support student use of the CER (Claim-Evidence-Reason) framework.

Expected From Students:

- Apply their understanding of sound movement.
- Use evidence from activities to explain how sound travels.
- Reflect on how we can use sound to solve real problems.

Parts of the book included.

Hold the spoons in the air and tap it gently. Listen carefully.
What do you hear?

Screen Lessons
Let's watch a video to see:
1. How your voice makes sound?
2. Why sound needs air to travel?
3. How ears catch sound waves?
<https://www.youtube.com/watch?v=gdGyyGP21GQ>

Assessment

Apprentice
1. Where CANT sound travel?
☐ Empty space.
☐ Water.

2. What happens when you hit a drum?
☐ It wiggles fast.
☐ It glows

STEAM SCOUTS

Builder
Explore How Sound Changes!
• Take two flat sticks and one rubber band.
• Stretch the rubber band around the sticks so there's a small space between them.
• Blow gently through the space and listen to the sound.
• Now, tighten the rubber band a little (make it shorter) and blow again.
• Then, loosen the rubber band a little (make it longer) and listen to how the sound changes.

Figure 11: A simple sound maker.

CER

Claim
What do you think will happen?
I think the sound is made by the _____.

Evidence
What did you observe?
I observed that when the rubber band is tight and short, the sound is _____, and when the rubber band is loose and long, the sound is _____.

Reasoning
Why did it happen?
I think the sound changes because the rubber band vibrates _____ when it is short and _____ when it is long.
This is like a real instrument or someone's voice because both use _____ to make sound.

STEAM SCOUTS

Hey friends, let's wrap up!

What is Sound?
Sound happens when something moves back and forth. This is called a vibration.

How Does Sound Move?
Sound travels in waves. These waves move through air, water, and solids (like a table or a spoon).

Where Does Sound Travel Best?
Sound moves best through solids, like wood or metal. It moves slower in water, and slowest in air.

What Happens When You Change the Rubber Band's Length?
If the rubber band is short, the sound is high. If the rubber band is long, the sound is low.

Mastermind
Make your own guitar with rubber bands! Try different lengths and see how the sound changes when you pluck them.

Figure 12: A simple string guitar.

You are doing a great job, friends!

Now let's share our ideas and choose what we can make to help all kids communicate.

Now I Can
Say how sound is made and how it moves.
Tell how shaking (vibrations) makes sound.
Find out how sound moves through solids, liquids, and air.
Talk about how sound helps people in real life.
Show what happens when one thing causes another.

STEAM SCOUTS



The sound changes because the rubber band vibrates faster when it is tight and slower when it is loose. Faster vibrations make a higher pitch, and slower vibrations make a lower pitch. This is the same way real instruments work, because musical sounds are made by vibrations.



Apprentice

Objective:

Students will apply their understanding of sound concepts such as vibration, sound travel through different materials, and the conditions necessary for sound to move.

Answers:

Assessment 

 **Apprentice**

1. Where CAN'T sound travel?

☐ Empty space.

☐ Water.



2. What happens when you hit a drum?

☐ It wiggles fast

☐ It glows





Possible Strategies for Implementation:

- **Read Aloud with Visual Cues:**
 - Use gestures (e.g., fingers vibrating, holding “no air” for space) to support students’ understanding.
- **Pair-Share Before Answering:**
 - Allow students to discuss each question in pairs before choosing an answer. Ask: “What makes you think that?”
- **Use Props If Available:**
 - Bring a drum, metal spoon, rubber band, or string to demonstrate real-time vibrations and material comparisons.
- **Interactive Review Game:**
 - Use hand signs to check each answer together.
- **Anchor Chart Recap:**
 - Revisit the Solid–Liquid–Gas sound travel chart to reinforce key ideas visually before answering.



Builder

Objective:

Students will explore how vibration and tension affect sound and pitch by changing the tightness of a rubber band and explaining their findings through a Claim-Evidence-Reasoning (CER) response.

Materials Needed:

- 2 flat sticks (popsicle sticks)
- 1 rubber band
- 1 small strip of paper (or a piece of straw / folded paper) to act as a spacer
- Tape (optional, to hold the spacer in place)
- Student CER worksheet
- Pencil

Activity Instructions:

1. Build the Sound Maker
Place the two sticks on top of each other.
2. Put the paper strip between the sticks in the center to create a small space.
3. Wrap the rubber band around both ends of the sticks to hold them together and keep the paper spacer in place.
4. Make a Sound
Blow gently through the space where the paper spacer is.
5. Listen to the sound the rubber band makes as it vibrates.
6. Experiment
Tighten the rubber band by twisting it or moving it inward so it stretches more. Blow again and listen.
7. Loosen the rubber band by pulling it outward so it stretches less. Blow again and listen.
8. Observe and Compare
Ask: How did the sound change when the rubber band was tight versus loose?
9. Complete the CER Worksheet
Students write their Claim, Evidence, and Reasoning using their observations.

Teacher Notes :

- Spacer purpose: It creates space for air to move, which helps the rubber band vibrate and produce sound.
- Key vocabulary: vibration, pitch, high, low, tension
- Main concept: Pitch depends on vibration speed.
- Fast = High pitch | Slow = Low pitch



Mastermind

Objective:

Students will explore how sound changes based on rubber band length and tightness by building a simple rubber band guitar and experimenting with pitch.

Materials Needed:

- 1 empty tissue box, shoebox, or small cardboard container (open top works best)
- Assorted rubber bands (different thicknesses and lengths)
- Optional:
 - Pencil or cardboard tube (to raise the rubber bands like a bridge)
 - Tape (to secure the bands or label them)
 - Ruler (for measuring band stretch)
 - Markers/stickers (to decorate the guitar)

Extra Teacher Resource (Demo Video):

<https://www.tiktok.com/@happytoddlerplaytime/video/7069568181270023429>



Use this video to preview the activity setup and see how kids interact with rubber band guitars in action.

Activity Instructions (To Be Done at Home – Student Creativity Focused)

1. Build Your Own Guitar!

- Use any box you have at home—like a tissue box or small container—as the guitar body.
- Stretch rubber bands of different sizes across the opening to make “strings.”

2. Play and Explore:

- Try plucking each rubber band.
- Listen carefully:
 - Which band makes a higher sound?
 - What happens when you pull it tighter or use a thinner band?

3. Be Creative:

- Decorate your guitar using markers or stickers.

- You can also name your instrument and create a cool design!

4. Music Meets Science:

- Try playing a rhythm or making a short sound pattern.
- Think: How is this like a real guitar or musical instrument?

There are no rules—just experiment, explore sound, and have fun!



Now I Can

Objective:

Students will reflect on their learning by identifying key concepts they've mastered related to sound movement, vibrations, real-world applications, and cause-effect relationships.

Activity Instructions (For Teachers):

1. Introduce the Checklist:

- Say: "Let's think back on everything we learned about sound today—how it moves, how it helps people, and how we explored it in class."

2. Guide the Students Through the Items:

- Read each "Now I Can" statement aloud one at a time.
- Ask: "Can you give a thumbs up if you feel proud of this learning?"
- Encourage brief discussion or examples after each one.

3. Have Students Complete Their Checklists:

- Let students place a checkmark next to what they feel confident about.
- Remind them: "It's okay if you don't check everything—we're all still learning!"

4. Optional Extension:

- Ask students to circle the one they're most proud of.
- Let them draw or write about it in a sentence at the bottom of the page.

Relation to Standards

1. English Language Arts (ELA):

CCSS.ELA-LITERACY.SL.2.1

Participate in collaborative conversations with diverse partners about grade 2 topics and texts.

- Students discuss sound observations and share ideas about how sound helps communication.

CCSS.ELA-LITERACY.SL.2.2

Recount or describe key ideas from a text or information presented orally or through other media.

- Students reflect on video lessons and hands-on sound activities.

CCSS.ELA-LITERACY.W.2.8

Recall information from experiences or gather information from provided sources to answer a question.

- Students write CER explanations and answer sound-related questions.

2. Next Generation Science Standards (NGSS):

2-PS1-4

Construct an argument with evidence that some changes caused by heating or cooling can be reversed and some cannot. (supporting cause-effect logic)

3-PS2-2 (extended)

Make observations and measurements of an object's motion to provide evidence of a pattern.

- Students observe how vibrating objects (like rubber bands, spoons, or drums) create sound.

3-PS2-4

Plan and conduct an investigation to provide evidence that sound is produced by vibrating materials.

- Explored through spoon tapping, rubber band guitar, and water vibration demo.

3. ISTE Standards for Students (Technology):

Empowered Learner (1a, 1c):

- Students use digital tools (videos, models) to understand real-world problems.
- They explore how technology can help people communicate using sound.

Innovative Designer (4a, 4b):

- Students build models and test solutions (e.g., rubber band instruments, sound challenges).
- They learn how communication tools work and how to design their own.